A timetable that works

Optimization · Level 3 · Lesson 1



Today we will

- Write a timetable as a model: a period per lesson, and rules.
- See how a solver crosses out and searches, and what INFEASIBLE means.
- Judge a timetable that keeps every rule: is it also fair?

Remember

- Periods are variables; each rule is one line, such as add_all_different for a class or a teacher.
- The solver crosses out what can’t work, then guesses and undoes dead ends.
- INFEASIBLE proves the rules clash; OPTIMAL doesn’t prove the timetable is fair.

The sticky-note timetable

1. Warm-up (5 min): Who has ever had two lessons at once?
2. Teach (8 min): The two rules and the extra rules.
3. Build it (15 min): Groups place sticky notes on the grid.
4. New rule (7 min): Add one rule and count what moves.
5. Wrap-up (5 min): Who got the last period? Is that fair?

Prepare twelve notes per group: four lessons for each of 5A, 5B and 6A (maths, science, English, art), one colour per subject. No sticky notes? Squares of paper and a little tape work.

Let’s discuss

1. Which rule did we forget at first, and how did we notice?
2. What does it mean when no timetable can keep every rule?
3. Who got the last period, and is that fair?

Exit ticket

- Write one rule of our school’s timetable, and one change that would make it fairer.